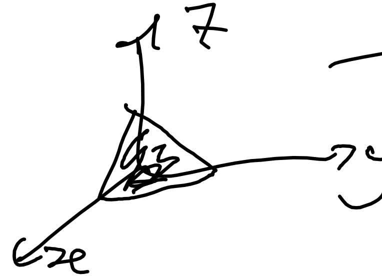
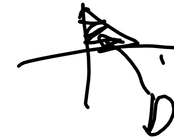
$(x, y) \in D, \quad g_1(x,y) \leq z \leq g_2(x,y)$

$$\iint_D \left(\int_{g_1(x,y)}^{g_2(x,y)} f(x,y,z)\, dz \right) dx\,dy$$

for D think of your top-down view

e.g. $\iiint_R (x+z)\,dx\,dy\,dz$





$R = \{(x,y,z)\in\mathbb{R}^3 : x\geq 0,\ y\geq 0,\ z\geq 0,\ x+y+z\leq 1\}$

$(x,y)\in D, \quad 0\leq z\leq 1-x-y$

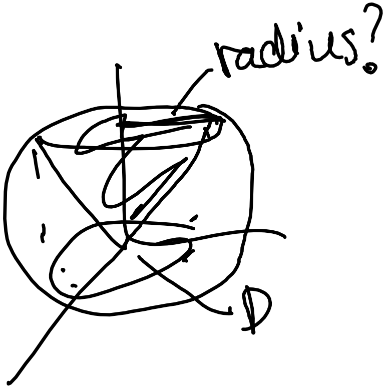
$D: x\geq 0, \quad 0\leq y\leq 1-x$

$$\iiint_R = \iint_D\left(\int_0^{1-x-y} x+z\,dz\right)dx\,dy = \iint_D \left[xz+\frac{z^2}{2}\right]_0^{1-x-y} dx\,dy$$

$$= \int_0^1\int_0^{1-x} x(1-x-y)+\frac{1}{2}(1-x-y)^2\,dy\,dx$$

DIY from here

recall the example I started yesterday



$$D = \{(x,y) \in \mathbb{R} : x^2+y^2 \leq ? \}$$

$$R = \{ (x,y,z) \in \mathbb{R}^3 : (x,y) \in D \subset \mathbb{R}^2, \sqrt{x^2+y^2} \leq z \leq \sqrt{1-x^2-y^2} \}$$

where intersect? $z = \sqrt{x^2+y^2}$ $\qquad x^2+y^2+z^2 = 1$

$z^2 = |x^2+y^2| = x^2+y^2$ $\qquad$ so $x^2+y^2+x^2+y^2=1$

so $2(x^2+y^2)=1 \Rightarrow x^2+y^2 = \frac{1}{2}$ so radius is $\frac{1}{\sqrt{2}}$

thus D: $x^2+y^2 \leq \frac{1}{2}$.

$$\iint_D \left(\int_{\sqrt{x^2+y^2}}^{\sqrt{1-x^2-y^2}} x \, dz \right) dx\,dy = \iint_D x \left(\sqrt{1-x^2-y^2} - \sqrt{x^2-y^2}\right) dx\,dy$$

$$\iint_D x\left(\sqrt{1-x^2-y^2} - \sqrt{x^2-y^2}\right) dx\,dy$$

now, we know D, and we know polar may work better!

$0 \le r \le \frac{1}{\sqrt{2}}, \quad 0 \le \theta \le 2\pi$

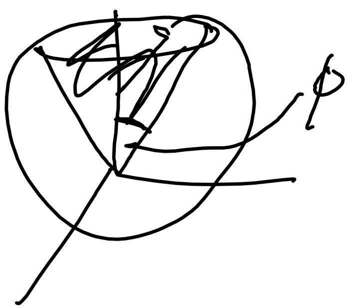
$$\int_0^{2\pi}\int_0^{1/\sqrt{2}} r\cos\theta \left(\sqrt{1-r^2} - \sqrt{r^2}\right) r\,dr\,d\theta$$

could we do change of variables from the start?

let's think of this conic cut of the sphere in spherical.

Recall $x = \rho\cos\theta\sin\phi$
$y = \rho\sin\theta\sin\phi$
$z = \rho\cos\phi$

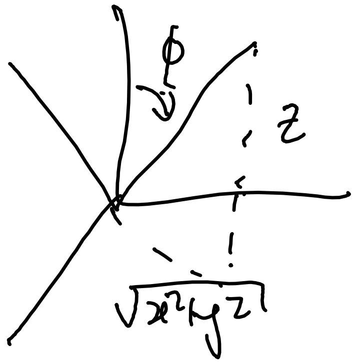
$\theta \in [0, 2\pi]$
$\phi \in [0, \pi]$
$\rho \in [0, \infty)$



radius of sphere? 1 so $0 \le \rho \le 1$
ϕ is $\frac{\pi}{4}$ in this case so $0 \le \phi \le \frac{\pi}{4}$
θ all the way so $0 \le \theta \le 2\pi$

$$\int_0^{2\pi}\int_0^{\pi/4}\int_0^{1} f\,|J|\,d\rho\,d\phi\,d\theta$$ nice!

what if cone was $z = \sqrt{3(x^2+y^2)}$



$$\frac{z}{\sqrt{x^2+y^2}} = \sqrt{3}$$

$$\tan\phi = \frac{\sqrt{x^2+y^2}}{z} = \frac{1}{\sqrt{3}}$$

so $\phi = \arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}$

so $0 \le \rho \le 1$ still in sphere

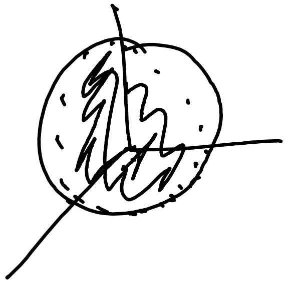
$0 \le \theta \le 2\pi$ still all around

$0 \le \phi \le \frac{\pi}{6}$

$$\int_0^{\pi/6}\int_0^{2\pi}\int_0^1 f\,|J|\; d\rho\, d\theta\, d\phi$$

What about

$x \geq 0 \quad y \geq 0$ in sphere w/ radius 2?



$0 \leq \rho \leq 2$

$0 \leq \theta \leq \frac{\pi}{2}$

$0 \leq \phi \leq \pi$

---

or

$x \leq 0 \quad y \leq 0$?

$z \leq 0$?

$0 \leq \rho \leq 2$

$\pi \leq \theta \leq \frac{3\pi}{2}$

$\frac{\pi}{2} \leq \phi \leq \pi$

let's do a cylindrical

$x = r\cos\theta$
$y = r\sin\theta$
$z = z$

R: $x^2+y^2 \le 1$ and $0 \le z \le 3\sqrt{x^2+y^2}$

$0 \le r \le 1$ $\quad 0 \le \theta \le 2\pi$

$0 \le z \le 3r$

think about this a bit

so $$\int_0^{2\pi}\int_0^1\int_0^{3r} f(r\cos\theta, r\sin\theta)\,|J|\,dz\,dr\,d\theta$$

change of variables does not necessarily mean you get a "box" domain! check 3(ii) in tut 11 for double int. where change leads to non-rectangular

Jacobians?

cylindrical: $x = r\cos\theta \quad y = r\sin\theta \quad z = z$

$$J = \begin{vmatrix} x_r & x_\theta & x_z \\ y_r & y_\theta & y_z \\ z_r & z_\theta & z_z \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \cos\theta\,(r\cos\theta) - -r\sin\theta(\sin\theta) + 0$$

$$= r\cos^2\theta + r\sin^2\theta = r$$

$$|J_{cylinder}| = r$$

nice.

Spherical:

$$x = \rho \cos\theta \sin\phi$$
$$y = \rho \sin\theta \sin\phi$$
$$z = \rho \cos\phi$$

$$J = \begin{vmatrix} x_\rho & x_\theta & x_\phi \\ y_\rho & y_\theta & y_\phi \\ z_\rho & z_\theta & z_\phi \end{vmatrix} =$$

$$\begin{vmatrix} \cos\theta\sin\phi & -\rho\sin\theta\sin\phi & \rho\cos\theta\cos\phi \\ \sin\theta\sin\phi & \rho\cos\theta\sin\phi & \rho\sin\theta\sin\phi \\ \cos\phi & 0 & -\rho\sin\phi \end{vmatrix} = -\rho^2\sin\phi$$

$$|J_{spher}| = \rho^2 \sin\phi$$